

EE-482: Network Security and Cryptography

LECTURES NO 12: *Digital Signatures*

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Organization of Lecture

1. Asymmetric Key Cryptography
2. Disputes between two parties
3. Digital Signatures
4. Requirements of Digital Signatures
5. One-way Hash Function
6. Simple Hash Function

1. Asymmetric Key Cryptography (1/2)

Asymmetric Key Cryptography

Sender A

Receiver B

A encrypts message using B's public key

B decrypts message its private key

→ Prevents from interception attack

→ Provides confidentiality

↳ Consider another scheme:

Sender A; receiver B.

A encrypts the message using its private key

B decrypts the message using A's public key

1. Asymmetric Key Cryptography (2/2)

Q: What is the advantage of such a scheme?
↳ A's public key is known any party
can decrypt using A's public key.

Intention is not to hide the contents of message.

Intention is to achieve:

- ⇒ **Authentication:** verify the source of message
- ⇒ **Non-repudiation:** prevent sender/receiver from denying sending/receiving a message.

2. Disputes between two parties

Disputes between two parties

Example Scenario:

Alice sends a message to Bob



- 1) Bob may forge a message & claim it came from Alice (Authentication)
 - 2) Alice can deny sending the message
 - 3) Bob can deny receiving the message
- } (Non-repudiation)

To resolve such disputes Digital Signatures

3. Digital Signatures

* Digital Signatures

↳ Idea is similar to conventional signature on paper.

↳ Only the person that sends the message is capable of generating a valid signature

Digital Signatures must exhibit following properties:

1) **Authentic:**

Can verify who signed message.

2) **Unforgeable:**

Only signer could have signed message.

3) **Not Reusable:**

Cannot bind signature to different message.

4) **Unalterable:**

Cannot alter message without affecting its signature.

5) **Non-Repudiation**

Signer cannot later deny signing

4. Requirement of Digital Signatures

Requirement of Digital Signatures :

- 1) Sender dependant (unique to sender)
- 2) Easy to produce
- 3) Easy to verify
- 4) Infeasible to forge
- 5) Message dependant
- 6) Useful if signature can be stored separately (from signed message)

5. One-way Hash Function (1/5)

↳ also called Secure Hash Function or Message Digests

↳ Message digest is a fingerprint or summary of a message.

↳ Similar to the concept of a CRC (Cyclic Redundancy Check)

Original data

11100100	11011101	00111001	00101001
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Arrange as rows

1	1	1	0	0	1	0	0
1	1	0	1	1	1	0	1
0	0	1	1	1	0	0	1
0	0	1	0	1	0	0	1
0	0	1	0	1	0	0	1

XOR($\oplus$)

Append to original message

5. One-way Hash Function (2/5)

Message digest is based on same principle e.g

message: 4000

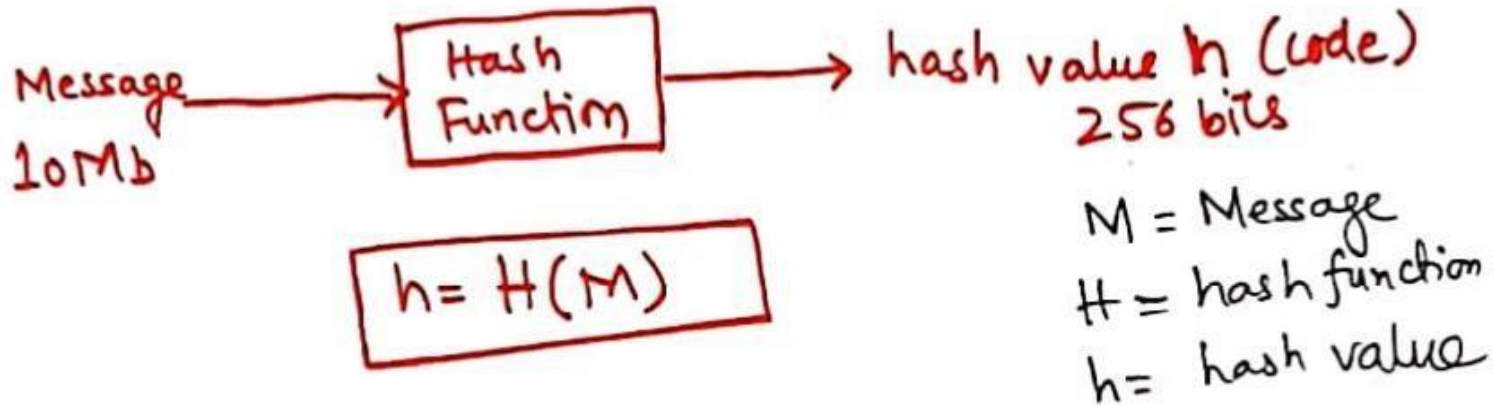
divide by 4 to get 1000

4: fingerprint of 4000

- Divide 4000 by 4 to get 1000 always.
- If change either 4000 or 4 the result is not 1000
- Given only 4 we cannot trace back equation
 $4 \times 1000 = 4000$

5. One-way Hash Function (3/5)

* Perform a hashing operation (or message digest algorithm) over a block of data to produce its hash/message digest
Smaller in size than original message



↳ Hash function must have following properties:

- 1) 'H' can be applied to any block of data.
- 2) 'H' produces a fixed length output.
- 3) Easy to produce 'h' from 'M'.
- 4) Size of 'M' >>> size of 'h' compression.

5. One-way Hash Function (4/5)

5) For any given code 'h' it is computationally infeasible to find 'M' such that :

$$H(M) = h ; \text{ One way function}$$

6) Given two messages, if we calculate their message digest, the two message digests must be different.

{ if two messages produce same message digest it is called a COLLISION }

Given one msg. 'M1' and digest 'h' it is not feasible to find another message 'M2' which will produce same message digest exactly bit by bit.



Append signature to the message

Function: Hash

5. One-way Hash Function (5/5)

Algorithms:

- MD5 (message digest version 5)
- SHA (secure hash algorithm)
- Hash functions are KEYLESS
- 'MD' or 'h' is only a fingerprint of the original message.
- no cryptographic function is used.

6. Simple Hash Function (1/2)

- Input (message) is viewed as a sequence of n -bit blocks.
- Input is processed one block at a time
- Simplest hash function: Bit by bit Exclusive-OR (XOR).

	Bit 1	Bit 2	Bit 3	Bit n
Block 1	b_{11}	b_{21}	b_{31}	b_{n1}
Block 2	b_{12}	b_{22}	b_{32}	b_{n2}
⋮				
Block m	b_{1m}	b_{2m}	b_{3m}	b_{nm}
Hash code	C_1	C_2	C_3	C_n

6. Simple Hash Function (2/2)

$$C_i = b_{i1} \oplus b_{i2} \oplus b_{i3} \oplus \dots \oplus b_{im}$$

m = number of n -bit blocks

b_{ij} = i th bit in j block

$\oplus$ = XOR operation

Append hash code to the message as digital signature.

$$C_1 = b_{11} \oplus b_{12} \oplus b_{13} \oplus \dots \oplus b_{1m}$$

References: Textbooks

1. *“Cryptography and Network Security”, by Behrouz A. Forouzan, Debdeep Mukhopadhyay, 2nd Edition, McGraw Hill Education (India) Private Limited.*
2. *“Cryptography and Network Security Principles and Practice”, by William Stallings, 6th Edition, Pearson.*
3. *“Computer Security Principles and Practice”, by William Stallings, Lawrie Brown, 4th Edition, Pearson.*